

Navigators IDR — Implementation Plans

Phase 1: Architecture Recovery (Completed)

- **Goal:** Invalidate false benchmarks and establish a scientifically sound benchmark script.
- **Outcome:** `benchmark.py` rewritten to remove leakage and explicitly require a trained ML model.

Phase 2: Data Pipeline & Training (Completed)

- **Goal:** Generate datasets and train the TCN.
- **Outcome:** Created `dataset_builder.py` and `data_loader.py`. Trained TCN and saved `norm_stats.json` for proper scaling.

Phase 3: ONNX Export & Integration (Completed)

- **Goal:** Ensure the AI model can run offline on edge devices (JavaScript target).
- **Outcome:** Rewrote `verify_onnx.py` to export the PyTorch model to `model.onnx` with parity testing. Updated `benchmark.py` and `server.py` to use proper normalizers and the trained model.

Phase 4: Frontend Translation (Pending)

- **Goal:** Move the 15-state EKF and ONNX inference from the Python dev backend to the JavaScript frontend.
- **Method:** Use `onnxruntime-web` for inference and implement the EKF matrix operations in JS/WASM.

Phase 5: Map Matching Integration (Pending)

- **Goal:** Replace the synthetic map grid with OpenStreetMap matching.
- **Method:** Use offline spatial trees to snap EKF outputs to known road geometries.